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ODD END SEMESTER EXAMINATION DEC 2025

Name of the Course: B.Tech.

Name of the paper: ENGINEERING MECHANICS

Time: 3 hours

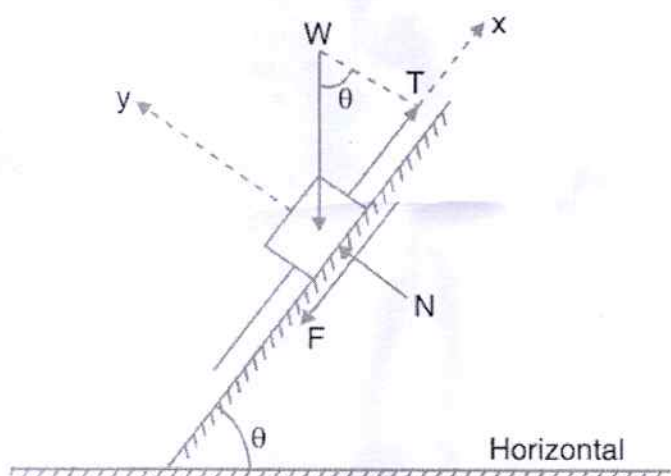
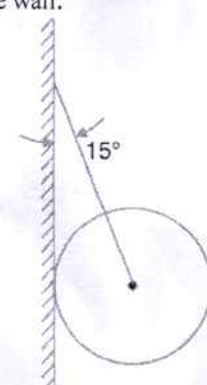
Semester: III

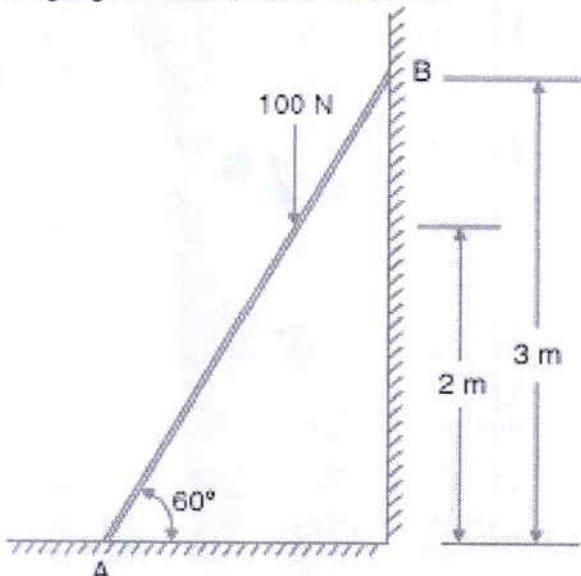
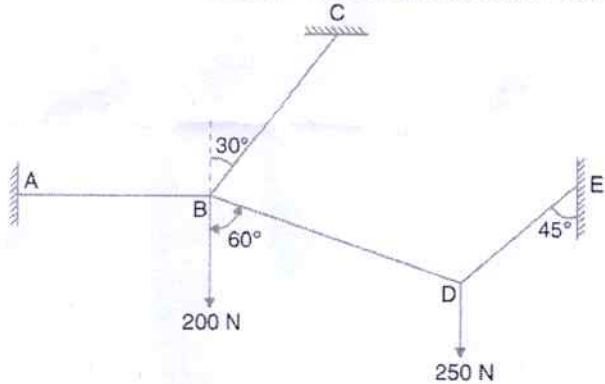
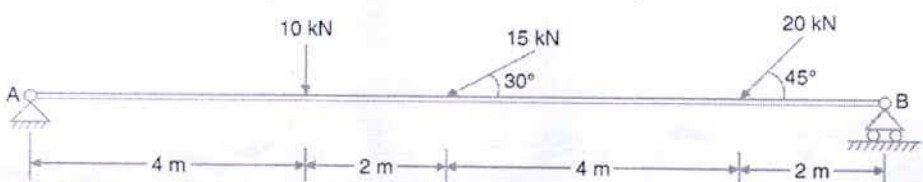
Paper Code: TME 306

Maximum Marks: 100

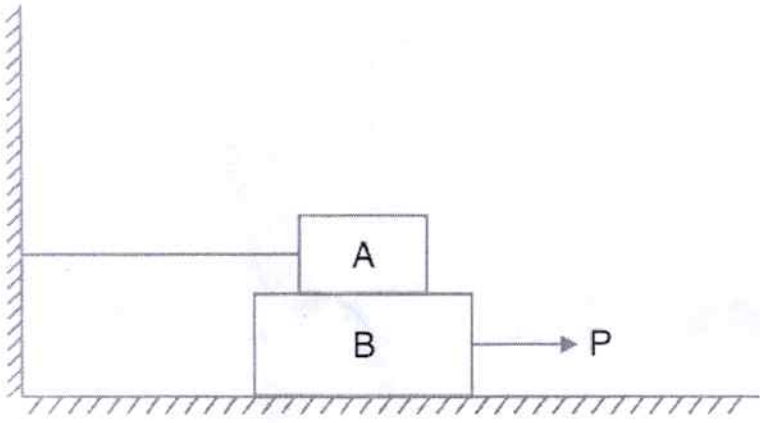
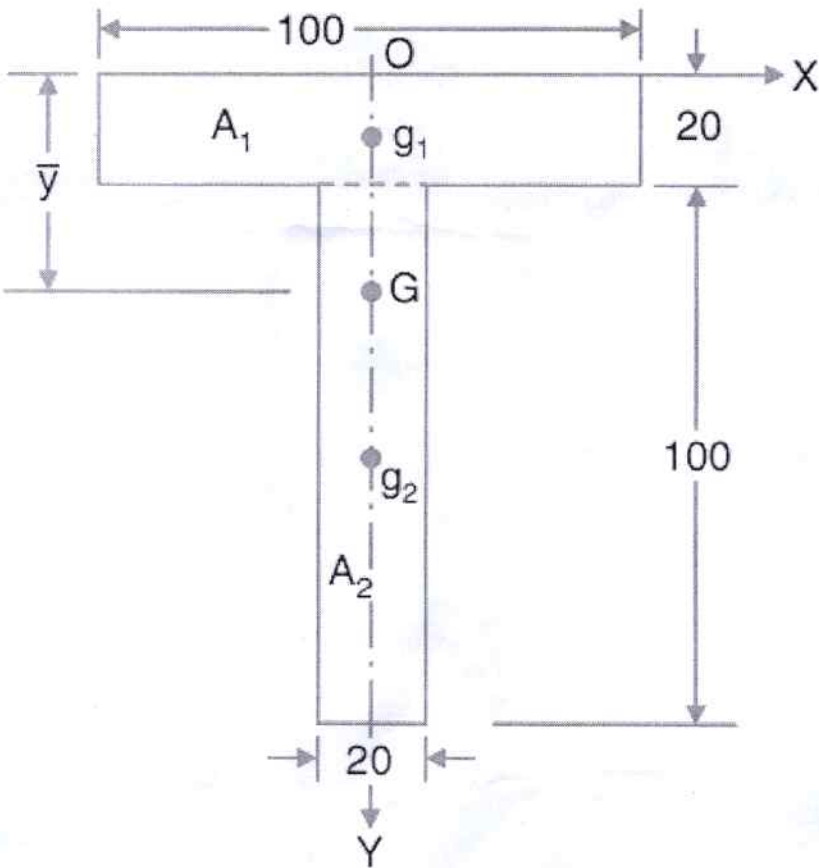
Note:-

- (i) All questions are compulsory
- (ii) Answer any two sub-questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	20 marks	
(a)	Differentiate between concurrent, non-concurrent, coplanar, and non-coplanar force systems with suitable examples.	
(b)	A system of forces acting on a body resting on an inclined plane is as shown in Fig. Determine the resultant force if $\theta = 60^\circ$, $W = 1000$ N, vertically downward, $N = 500$ newton acting normal to the plane, $F = 100$ N, acting down the plane and $T = 1200$ N, acting parallel to the plane. 	CO 1
(c)	A sphere weighing 100 N is tied to a smooth wall by a string as shown in Fig. Find the tension T in the string and the reaction R from the wall. 	

Q2	20 marks	
(a)	State Lami's theorem. Write the conditions under which it is applicable.	
(b)	A ladder weighing 100 N is to be kept in the position shown in Fig., resting on a smooth floor and leaning on a smooth wall. Determine the horizontal force required at floor level to prevent it from slipping when a man weighing 700 N is at 2 m above floor level. 	CO 2
(c)	A system of connected flexible cables, shown in Fig., is supporting two vertical forces 200 N and 250 N at points B and D. Determine the forces in various segments of the cable. 	
Q3	20 marks	
(a)	The beam AB of span 12 m shown in Fig. is hinged at A and is on rollers at B. Determine the reactions developed at A and B due to the loading shown in the figure. 	CO 3
(b)	Find the forces in all the members of the truss shown in Fig	

(c)	Determine the forces in the members FH, HG and GI in the truss shown in Fig. Each load is 10 kN and all triangles are equilateral with sides equal to 4 m.	
Q4	20 marks	
(a)	What is the value of P in the system shown in Fig. to cause the motion of a 500 N block to the right side? Assume the pulley is smooth and the coefficient of friction between other contact surfaces is 0.20.	CO 4
(b)	Block A weighing 1000 N rests over block B which weighs 2000 N as shown in Fig.. Block A is tied to the wall with a horizontal string. If the coefficient of friction between blocks A and B is 0.25 and between B and the floor is $\frac{1}{3}$, what should be the value of P to move the block (B).	

		
(c)	In a lifting machine, an effort of 500 N is to be moved by a distance of 20 m to raise a load of 10,000 N by a distance of 0.8 m. Determine the velocity ratio, mechanical advantage and efficiency of the machine. Determine also ideal effort, effort lost in friction, and the ideal load	
Q5		20 marks
(a)	Derive an expression to calculate the Centroid of a Triangle. Also, define moment of inertia and the principle axis theorem.	CO
(b)	Locate the centroid of the T-Section	5
	 All dimensions in mm	
(c)	Determine the moment of inertia of the L-section shown in the Fig. about its centroidal axes parallel to the legs. Also find the polar moment of inertia	

